

Sahasra Sprint 1-3 Detailed Project Report

Sports Biomechanics, Fatigue Prediction, Genomic-Inspired Recovery Clustering, and Workload Recommendation Prototype

Project Mode	Single-athlete longitudinal prototype
Primary Athlete	athlete_001
Sprint 1 Input	One athlete video
Sprint 2 Sessions	150 synthetic training sessions
Primary Dashboard Model	XGBoost
Readiness	Ready to continue to Sprint 4 as a prototype

Important interpretation: the current system analyzes one video-tracked athlete. The 150 rows shown in Sprint 2 and the dashboard are synthetic longitudinal sessions for athlete_001, not 150 different real athletes.

1. Executive Summary

The Sprint 1-3 pipeline is now connected end to end. Sprint 1 extracts lower-body biomechanics from an athlete video. Sprint 2 combines those biomechanics with wearable-style telemetry and trains fatigue risk models. Sprint 3 consumes Sprint 2 fatigue state, integrates genomic-inspired profile features, clusters recovery-performance phenotype states, and generates fatigue-aware workload recommendations.

Area	Current Implementation	Status
Sprint 1	MediaPipe pose extraction, 33 landmarks, knee/hip/ankle metrics, overlay video	Implemented
Sprint 2	150-session telemetry fusion, XGBoost, LSTM, ANN, dashboard	Implemented
Sprint 3	ACTN3/ACE profile, K-Means clustering, workload recommendations	Implemented
Integration	Sprint 2 fatigue outputs feed Sprint 3 for athlete_001	Connected
Limitations	Synthetic telemetry/genomics, no live sensor ingestion, no clinical injury labels	Prototype caveat

2. Expected Pipeline Flow

Expected flow: Sprint 1 video biomechanics -> Sprint 2 telemetry fusion and fatigue prediction -> Sprint 3 genomic-inspired recovery clustering -> fatigue-aware workload recommendations.

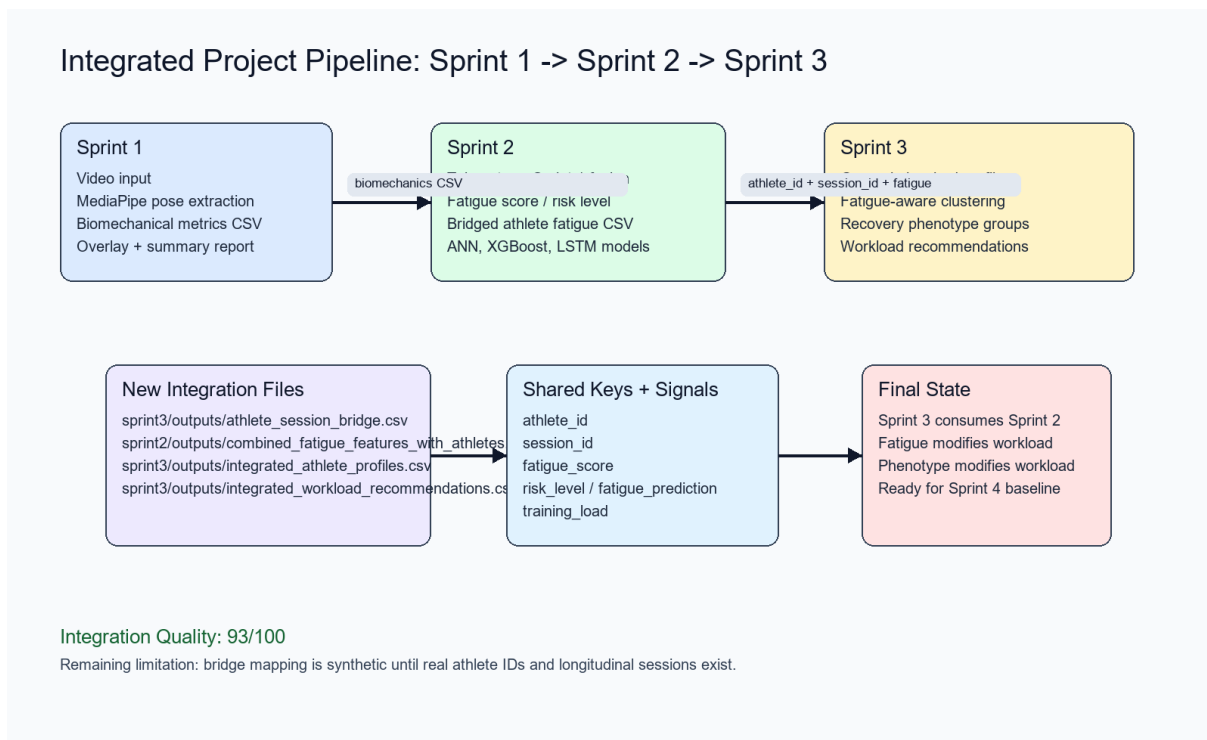


Figure 1. Integrated project pipeline diagram.

3. Sprint 1: Computer Vision and Biomechanics

Sprint 1 processes one athlete video using MediaPipe pose tracking. It extracts a 33-point body coordinate map and computes lower-body biomechanics over time.

Requirement	Implemented Evidence
Video ingestion	sprint1/pose_pipeline.py processes video frame by frame
33-point tracking	CSV has 148 columns including x/y/z/visibility for landmarks
Knee flexion	right_knee_angle, left_knee_angle, knee_angle_difference
Ankle deviation	right_ankle_deviation_angle, left_ankle_deviation_angle
Hip alignment	hip_alignment_angle, hip_alignment_abs_angle, hip_y_difference
Outputs	Enhanced CSV, annotated overlay MP4, biomechanical summary

Sprint 1 core question coverage

The Sprint 1 core question asks whether real-time geometric movement mechanics can isolate soft-tissue degradation patterns before clinical strain symptoms emerge. The project covers the movement-mechanics part by producing frame-level geometric indicators. It does not yet prove clinical soft-tissue degradation because verified strain labels are not available.

4. Sprint 2: Telemetry Fusion and Fatigue Prediction

Sprint 2 uses Sprint 1 biomechanics plus 150 synthetic wearable/workload sessions for athlete_001. The primary model for dashboard decision-making is XGBoost because it has an inspectable held-out split and feature importance.

Input/Output		Path or Description			
Telemetry input		sprint2/input_data/synthetic_athlete_telemetry_fresh.csv			
Combined features		sprint2/outputs/combined_fatigue_features.csv			
Athlete-linked features		sprint2/outputs/combined_fatigue_features_with_athletes.csv			
XGBoost model		sprint2/models/xgboost_fatigue_model.json			
Dashboard		dashboard/athlete_fatigue_dashboard.html			
Model	Accuracy	Precision	Recall	F1	Dashboard Use
XGBoost	0.7895	0.8288	0.7895	0.7844	Primary decision model
LSTM	0.7667	0.7782	0.7667	0.7714	Sequence comparison prototype
ANN	0.500	N/A	N/A	N/A	Weak baseline

Risk distribution

Risk Level	Session Count
High	76

Risk Level	Session Count
Low	40
Moderate	34

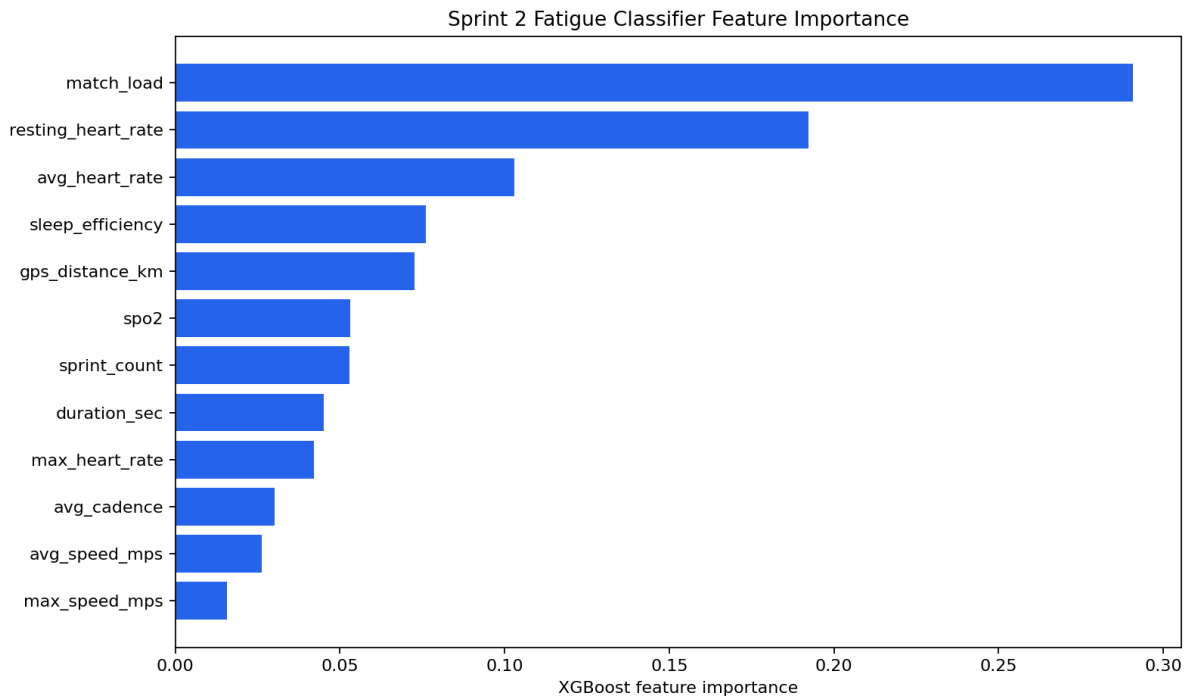


Figure 2. XGBoost feature importance chart.

Sprint 2 core question coverage

The Sprint 2 core question asks whether passive physiological streams and active video tracking can extract hidden fatigue variables better than individual analytics silos. The implemented prototype directly addresses this by combining wearable-style telemetry with Sprint 1 video biomechanics and comparing XGBoost, LSTM, and ANN models. Because the data is synthetic, this is workflow evidence rather than real-world validation.

5. Sprint 3: Genomic-Inspired Profiling and Workload Optimization

Sprint 3 simulates genomic-inspired athlete profiling using ACTN3 and ACE markers, recovery metrics, fatigue state, and training load. After the single-athlete correction, all 150 sessions belong to athlete_001.

Component	Implemented Evidence
Genomic context	ACTN3 and ACE genotype fields in sprint3/input_data/athlete_genomic_profiles.csv
Sprint 2 integration	sprint3/outputs/integrated_athlete_profiles.csv consumes fatigue_score and risk_level
K-Means clustering	sprint3/models/kmeans_recovery_model.pkl and clustered_athletes.csv
Workload scaling	workload_multiplier, workload_adjustment_percent, recommended_training_load
Safety warning	Reports state genomic markers are probabilistic and not deterministic limits

Phenotype and recovery outputs

Output Category	Distribution
Phenotype groups	{'High Recovery / Power': 84, 'Recovery Limited': 66}
Recovery priority	{'High': 142, 'Moderate': 8}

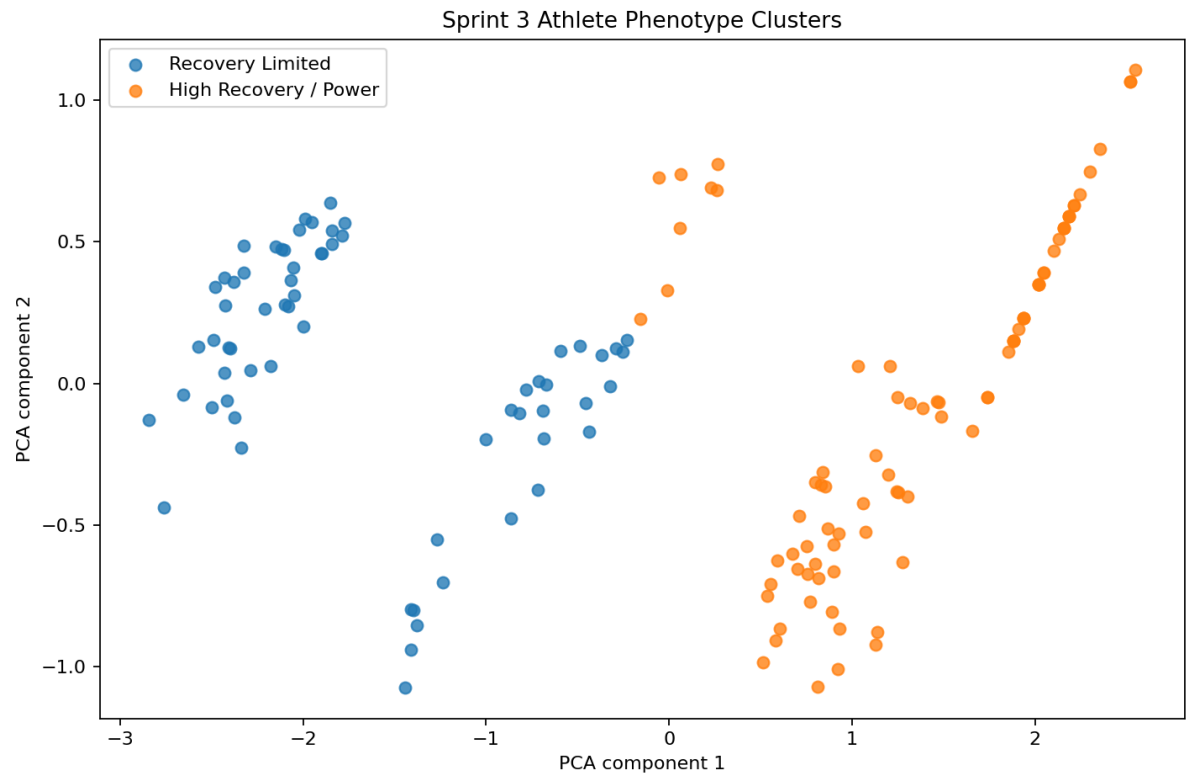


Figure 3. Sprint 3 recovery-performance cluster visualization.

Sprint 3 core question coverage

The Sprint 3 core question asks whether open literature genomic markers can calibrate athlete workloads safely without deterministic genetic bias. The prototype covers this by using ACTN3 and ACE as probabilistic tendency features and by keeping workload decisions tied to fatigue and recovery state. It does not yet use real genomic datasets or validated training ceiling models.

6. Dashboard Output

The dashboard is a static HTML interface for athlete_001. It shows session-level fatigue score, risk level, phenotype group, recovery priority, XGBoost prediction correctness, feature importance, and workload recommendation.

Dashboard Item	Current Output
HTML file	dashboard/athlete_fatigue_dashboard.html
Test predictions	dashboard/outputs/xgboost_test_predictions.csv
XGBoost held-out rows	38
Correct / wrong	30 correct, 8 wrong
Dashboard framing	Single Athlete Fatigue Dashboard

7. Missing or Prototype-Level Items

- True multi-angle video processing is not implemented.
- Cutting and jumping action examples are not demonstrated.
- Telemetry is synthetic, not live smartwatch/API data.
- Genomic data is synthetic/literature-inspired, not a real open-source genomic dataset.
- ACTN3 and ACE are categorical markers, not full polygenic sequence analysis.
- No clinically validated injury or strain outcome labels are available.
- Dashboard is static HTML, not real-time streaming.
- Personal training ceilings are approximated by workload scaling, not clinically validated ceiling prediction.

8. Final Readiness Decision

READY TO CONTINUE TO SPRINT 4 AS A PROTOTYPE. The implemented pipeline satisfies the main Sprint 1-3 engineering workflow and addresses the core research questions at prototype level. The remaining gaps should be presented as future work and validation requirements.